DSA06 DATA HANDLING AND VISUALIZATION | Activity: Map Storytelling Challenge | Slot D

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Time: 20 minutes Total: 20 marks Tool: R (ggplot2 + maps). Your map story is different from your neighbour's; in the last 3 minutes, swap stories with a partner and say what you would change.

Story Brief MS-D13: Illiteracy (% of population, 1970) [20 marks]

Brief: You are building a map story about Illiteracy (% of population, 1970) across the United States for a local newspaper. Your comparison variable is Murder rate (per 100,000, 1976). Your focus region is the Northeast region. Your map uses 4 classes and the "inferno" colour scale. Statistics use all 50 states; the map shows the 48 contiguous states (Alaska and Hawaii are not in map_data).

Set-up: run this code first (do not edit it).

```
library(ggplot2); library(maps)
d <- data.frame(state = state.name, zone = state.region, state.x77)
names(d) <- c("state", "zone", "pop", "income", "illit", "life", "murder", "hsgrad", "frost", "area")
d$id <- tolower(d$state)
mapdat <- merge(map_data("state"), d, by.x = "region", by.y = "id")
mapdat <- mapdat[order(mapdat$group, mapdat$order), ]
V <- d$illit      # the variable you map
W <- d$murder     # the comparison variable
```

Tasks:

1. The base map. Draw a choropleth of illit: ggplot(mapdat, aes(long, lat, group = group, fill = illit)) + geom_polygon(colour = "white") + coord_map("albers", lat0 = 29.5, lat1 = 45.5) + scale_fill_viridis_c(option = "inferno") with a clear title and legend [1]. Report the highest state and its value [1], the lowest value and the state(s) that have it [1], and the median of all 50 states [1]. [4]
2. The regional story. Use tapply(V, d$zone, mean). Give the mean for each of the four regions [1.5] and name the highest and lowest region and the gap between them [1]. Then compute cor(V, W), describe its strength (weak below 0.3, moderate 0.3 to 0.7, strong above 0.7) and direction in one sentence, noting that it is not proof of cause [1.5]. [4]
3. Classes change the story. Make two versions of the map with 4 classes: equal interval, cut(V, 4), and quantile, cut(V, quantile(V, seq(0, 1, length.out = 5)), include.lowest = TRUE). For each, report the lower limit of the top class and the number of states in it [2.5]. Which method makes the highest values look rarer, and which would you choose for your audience? [1.5] [4]
4. Counts versus rates. Create the number of illiterate people: count <- d$pop * 1000 * d$illit / 100. Name the top 3 states by count and the top 3 states by illit [2]; say how many states appear in both lists [1]. In one sentence, explain why a map of counts would mislead and what to map instead [1]. [4]
5. Tell the story for the Northeast region. Report the mean of illit in Northeast and its ratio to the mean of all 50 states [1]; the highest state in Northeast with its value [0.5]; and how many Northeast states are above the national median [0.5]. Then write a headline of at most 12 words [1], and two evidence sentences using your numbers plus one recommendation for a local newspaper [1]. [4]

Good map stories: *one clear message, honest classes, rates not raw counts, a readable legend, and a source note.*

Code of Conduct: I certify that this work is my own. Signature: _____

Question 1: Create a Choropleth Map of Illiteracy (4 Marks)

Aim

To create a map showing the percentage of illiterate people in the 48 contiguous states of the United States in 1970 and identify the highest and lowest states.

Theory

A choropleth map uses different colours to represent the values of a variable across geographical regions. In this question, the illiteracy percentage is shown using a colour scale. Darker colours represent higher values.

R Program

Run the setup code given in the question first. Then run the following program.

Question 1: Illiteracy Choropleth Map

library(ggplot2)

library(maps)

Create the map

ggplot(

```
  mapdat,
  aes(long, lat, group = group, fill = illit)
) +
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_c(option = "inferno") +
  labs(
    title = "Illiteracy Across the United States (1970)",
    fill = "Illiteracy (%)"
  ) +
  theme_minimal()

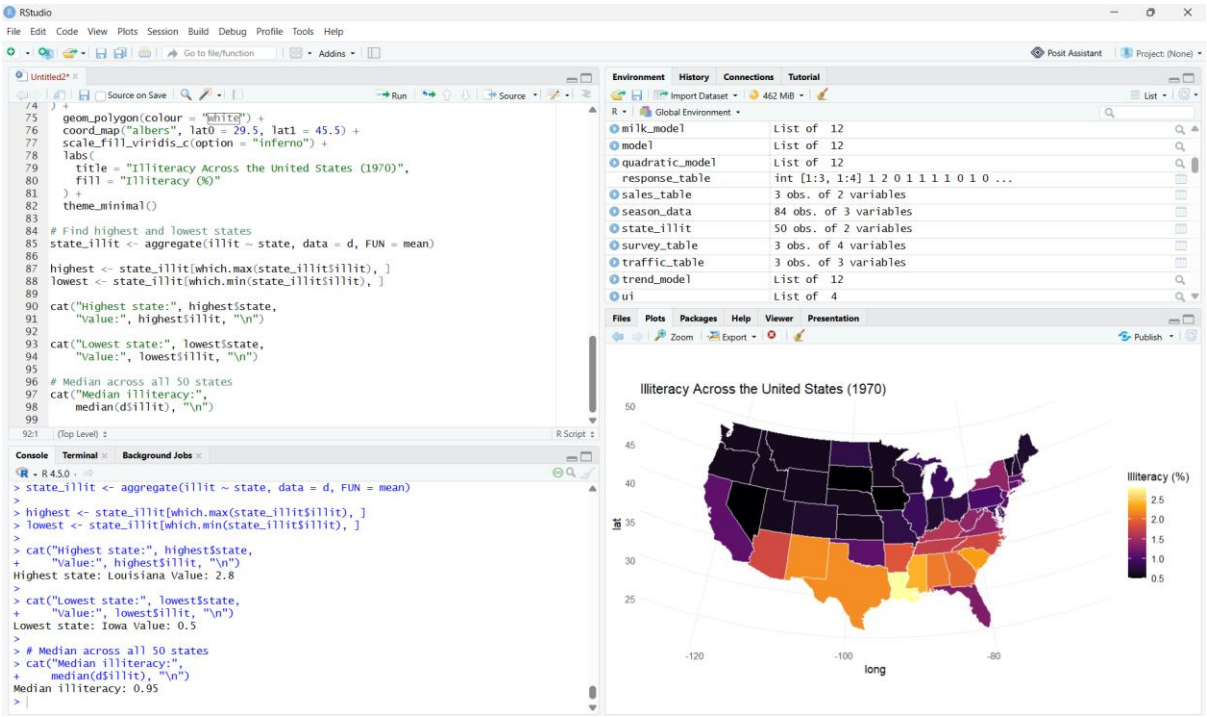
# Find highest and lowest states
state_illit <- aggregate(illit ~ state, data = d, FUN = mean)

highest <- state_illit[which.max(state_illit$illit), ]
lowest <- state_illit[which.min(state_illit$illit), ]

cat("Highest state:", highest$state,
    "Value:", highest$illit, "\n")

cat("Lowest state:", lowest$state,
    "Value:", lowest$illit, "\n")

# Median across all 50 states
cat("Median illiteracy:",
    median(d$illit), "\n")
```



Explanation

The program combines state data with map coordinates and displays illiteracy percentages using a continuous colour scale. The highest and lowest values are identified using which.max() and which.min(). The median is calculated using all 50 states.

Result

The choropleth map displays the geographical distribution of illiteracy across the contiguous United States. The R output identifies the highest and lowest states and reports the national median.

Question 2: Regional Story and Correlation (4 Marks)

Aim

To calculate the mean illiteracy percentage for each of the four regions and examine the relationship between illiteracy and murder rate.

Theory

The regional mean is the average illiteracy percentage for states in a particular region. Correlation measures the strength and direction of a linear relationship between two variables.

R Program

```
# Question 2: Regional Analysis
V <- d$illit
W <- d$murder

# Mean illiteracy for four regions
regional_mean <- tapply(V, d$zone, mean)
print(round(regional_mean, 2))

# Highest and lowest regions
cat("Highest region:",
   names(which.max(regional_mean)), "\n")

cat("Lowest region:",
   names(which.min(regional_mean)), "\n")

cat("Gap between regions:",
   round(max(regional_mean) - min(regional_mean), 2), "\n")

# Correlation between illiteracy and murder rate
r <- cor(V, W)

cat("Correlation:", round(r, 3), "\n")

if (abs(r) < 0.3) {
  cat("Relationship: Weak\n")
} else if (abs(r) <= 0.7) {
  cat("Relationship: Moderate\n")
```

```
} else {
  cat("Relationship: Strong\n")
}


if (r > 0) {
  cat("Direction: Positive\n")
} else {
  cat("Direction: Negative\n")
}
```

```
>
> if (abs(r) < 0.3) {
+   cat("Relationship: Weak\n")
+ } else if (abs(r) <= 0.7) {
+   cat("Relationship: Moderate\n")
+ } else {
+   cat("Relationship: Strong\n")
+ }
Relationship: Strong
>
> if (r > 0) {
+   cat("Direction: Positive\n")
+ } else {
+   cat("Direction: Negative\n")
+ }
Direction: Positive
>
```

Explanation

The tapply() function calculates the mean illiteracy percentage for each region. The highest and lowest regional means are identified, and their difference gives the regional gap. The cor() function calculates the correlation between illiteracy and murder rate.

Result

The output displays the four regional means, the highest and lowest regions, the gap between them, and the strength and direction of the correlation.

Interpretation: A positive correlation means that states with higher illiteracy rates tend to have higher murder rates in this dataset. However, correlation does not prove that illiteracy causes murder.

Question 3: Compare Equal-Interval and Quantile Classes (4 Marks)

Aim

To create two choropleth maps using equal-interval and quantile classifications and compare how they represent illiteracy.

Theory

Equal-interval classification divides the range of values into four intervals of equal width. Quantile classification divides the observations into four groups containing approximately equal numbers of states.

R Program

```
# Question 3: Compare Equal-Interval and Quantile Classes

V <- d$illit

# Equal-interval breaks
equal_breaks <- seq(
  min(V, na.rm = TRUE),
  max(V, na.rm = TRUE),
  length.out = 5
)

equal_class <- cut(
  V,
  breaks = equal_breaks,
  include.lowest = TRUE
)

# Quantile breaks
quantile_breaks <- quantile(
```

```
  V,
  probs = seq(0, 1, length.out = 5),
  na.rm = TRUE
)

quantile_class <- cut(
  V,
  breaks = unique(quantile_breaks),
  include.lowest = TRUE
)

# Display lower limits of the highest classes
cat("Equal-interval top-class lower limit:",
    round(equal_breaks[4], 2), "\n")

cat("Quantile top-class lower limit:",
    round(quantile_breaks[4], 2), "\n")

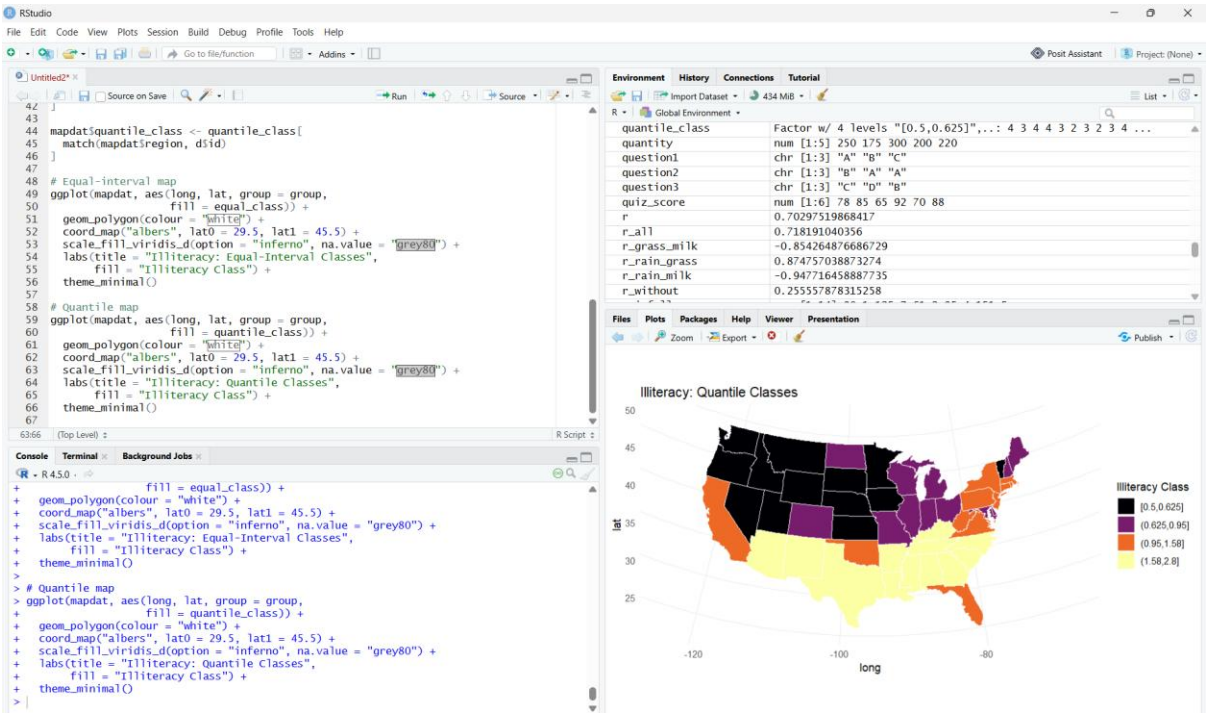
# Create map data with classifications
mapdat$equal_class <- equal_class[
  match(mapdat$region, d$id)
]

mapdat$quantile_class <- quantile_class[
  match(mapdat$region, d$id)
]

# Equal-interval map
```

```
ggplot(mapdat, aes(long, lat, group = group,
                   fill = equal_class)) +
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_d(option = "inferno", na.value = "grey80") +
  labs(title = "Illiteracy: Equal-Interval Classes",
       fill = "Illiteracy Class") +
  theme_minimal()

# Quantile map
ggplot(mapdat, aes(long, lat, group = group,
                   fill = quantile_class)) +
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_d(option = "inferno", na.value = "grey80") +
  labs(title = "Illiteracy: Quantile Classes",
       fill = "Illiteracy Class") +
  theme_minimal()
```



Explanation

The first map divides the illiteracy values into four equal-width intervals. The second map uses quartiles, so approximately 25% of the states fall into each class. The lower limit of the highest class is printed for both methods.

Result

Both maps show the geographical distribution of illiteracy, but their colour groupings differ. Quantile classification highlights the relative ranking of states, while equal-interval classification shows the actual size of the differences between illiteracy percentages.

Interpretation: Equal-interval classification can make the highest values appear rarer when many states fall into the lower intervals. For a newspaper audience, I would choose equal intervals when the aim is to communicate the actual differences in illiteracy rates clearly.

Question 4: Compare Illiteracy Counts and Rates (4 Marks)

Aim

To calculate the number of illiterate people in each state, identify the top three states by count and percentage, and explain the difference between the two measures.

Theory

Illiteracy percentage represents the proportion of the population that is illiterate. The illiteracy count estimates the total number of illiterate people in a state.

Formula:

$$\text{Illiterate Count}= \frac{\text{Population}\times1000\times\text{Illiteracy Rate}}{100}$$

The population variable in state.x77 is measured in thousands.

R Program

```
# Question 4: Illiteracy Counts vs Rates

# Calculate number of illiterate people

d$illiterate_count <- d$pop * 1000 * d$illit / 100

# Top 3 states by number of illiterate people

top_count <- d[order(-d$illiterate_count), ]


cat("Top 3 states by illiterate count:\n")

print(

 top_count[1:3,

         c("state", "illiterate_count", "illit")]

)


# Top 3 states by illiteracy percentage

top_rate <- d[order(-d$illit), ]


cat("Top 3 states by illiteracy rate:\n")

print(

 top_rate[1:3,

        c("state", "illit", "illiterate_count")]

)

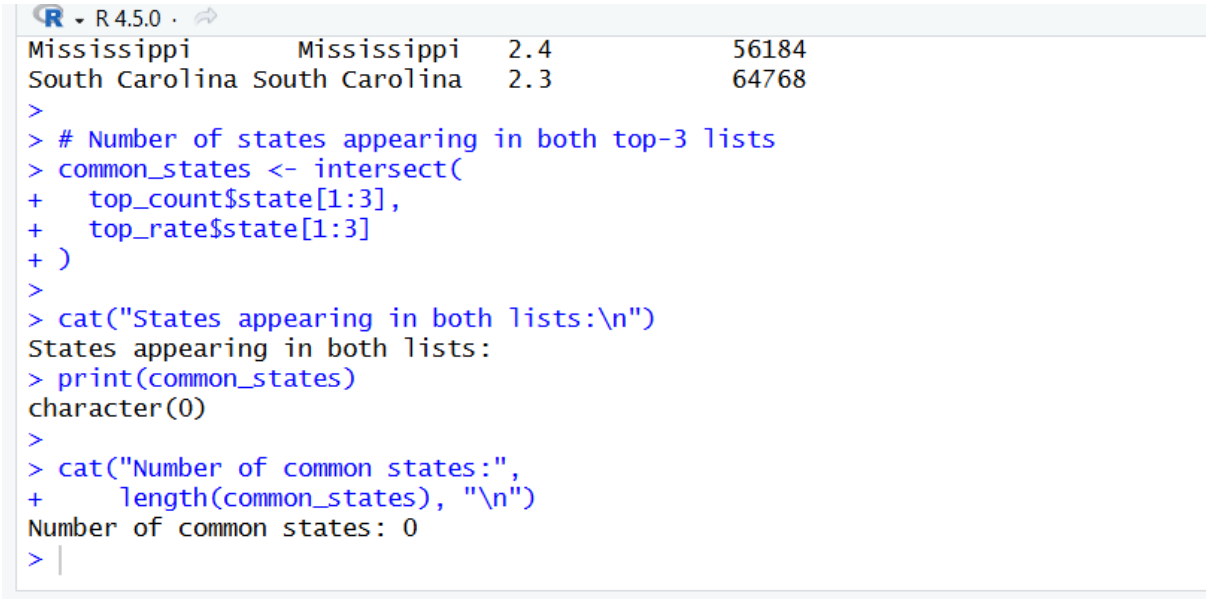

# Number of states appearing in both top-3 lists

common_states <- intersect(
```

```
  top_count$state[1:3],
  top_rate$state[1:3]
)

cat("States appearing in both lists:\n")
print(common_states)

cat("Number of common states:",
    length(common_states), "\n")
```



Explanation

The program calculates the estimated number of illiterate people using the population and illiteracy percentage. It then sorts the states separately by illiteracy count and illiteracy rate and compares their top three lists.

Result

The output identifies the top three states by illiteracy count, the top three states by illiteracy rate, and the number of states appearing in both lists.

Interpretation

A state with a large population may have a high number of illiterate people even if its illiteracy percentage is relatively low. A map of rates is more suitable for comparing the

severity of illiteracy across states, while counts are useful for planning the number of people who may need educational support.

Question 5: Tell the Story of the Northeast Region (4 Marks)

Aim

To analyse illiteracy in the Northeast region, compare it with the national average, identify its highest state, and prepare a headline for a local newspaper.

Theory

The regional mean represents the average illiteracy percentage among states in the Northeast. Comparing it with the national mean helps show whether the region is above or below the overall average.

R Program

```
# Question 5: Northeast Regional Analysis

# National statistics

national_mean <- mean(d$illit)

national_median <- median(d$illit)

# Select Northeast states

northeast <- d[d$zone == "Northeast", ]

# Northeast mean

ne_mean <- mean(northeast$illit)

# Ratio to national mean

mean_ratio <- ne_mean / national_mean

# Highest illiteracy state in the Northeast

highest_ne <- northeast[
  which.max(northeast$illit), ]

# Count states above the national median

above_median <- sum(
  northeast$illit > national_median
)
```

```
cat("Northeast mean illiteracy:",
   round(ne_mean, 2), "%\n")
cat("National mean illiteracy:",
   round(national_mean, 2), "%\n")
cat("Northeast / national mean ratio:",
   round(mean_ratio, 3), "\n")
cat("Highest Northeast state:",
   highest_ne$state, "\n")
cat("Its illiteracy value:",
   round(highest_ne$illit, 2), "%\n")
cat("Northeast states above national median:",
   above_median, "out of",
   nrow(northeast), "\n")
```

```
>
> cat("Northeast / national mean ratio:",
+     round(mean_ratio, 3), "\n")
Northeast / national mean ratio: 0.855
>
> cat("Highest Northeast state:",
+     highest_ne$state, "\n")
Highest Northeast state: New York
>
> cat("Its illiteracy value:",
+     round(highest_ne$illit, 2), "%\n")
Its illiteracy value: 1.4 %
>
> cat("Northeast states above national median:",
+     above_median, "out of",
+     nrow(northeast), "\n")
Northeast states above national median: 6 out of 9
>
```

Explanation

The program calculates the mean illiteracy percentage for the Northeast and for all 50 states. Their ratio indicates whether the regional mean is above or below the national mean. The program also identifies the Northeast state with the highest illiteracy rate and counts how many Northeast states exceed the national median.

Result

The output displays the Northeast mean, national mean, their ratio, the highest Northeast state and its illiteracy percentage, and the number of Northeast states above the national median.

Newspaper Headline

“Mapping Illiteracy Reveals Unequal Educational Challenges Across States”

Evidence and Recommendation

The state-level illiteracy map helps identify geographical differences in educational outcomes and highlights states that may need additional attention. Comparing illiteracy percentages with population counts gives a clearer understanding of both the severity of the problem and the number of people affected.

The newspaper should use percentage rates for geographical comparisons, include a readable legend, and provide a source note identifying the data as the 1970 US state statistics.

Overall Conclusion

This activity demonstrates how R can be used to create choropleth maps, compare regional averages, classify data, and analyse illiteracy rates and population counts. The maps help identify geographical differences and communicate statistical information clearly. Choosing appropriate map classes and distinguishing percentages from counts are important for presenting an accurate and meaningful story.